Statistical Properties of Ordinal Pattern Features and Their Application to Clustering Convex Combinations of Deterministic, Chaotic, and Stochastic Time Series

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1 Purpose

Ordinal pattern features are used to analyse simulated time series data generated from deterministic sine, ARMA, AR, MA, and logistic map models. The purpose of this study is to explore additional points in the entropy–complexity (HC) plane that were not identified in our previous experiments. In the earlier experimental design, the resulting points were mostly concentrated near the right-hand corner of the HC plane (Figure 1 shows it clearly), which made them difficult to use effectively for time series clustering. Therefore, we broaden the experimental design in order to generate HC points that are more widely distributed across the HC plane. This approach allows for improved visualization and a better representation of entropy and complexity values throughout the plane.

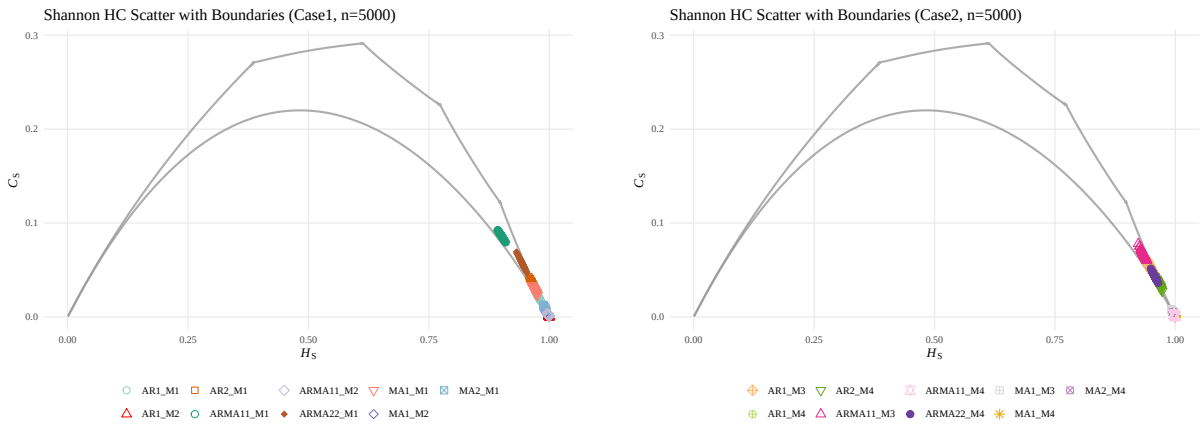

Figure 1: Entropy–complexity plane for Shannon entropy.

Once a sufficiently wide coverage of points in the HC plane is obtained, these data will be used to further compute additional entropy and complexity–related features. In particular, measures such as Rényi entropy, Tsallis entropy, and Fisher information will be calculated. These features will then be used to construct a multinomial logistic model in order to evaluate their ability to distinguish between different time series models. Model validation will be performed to identify the most informative features.

Finally, the validated features will be used for time series clustering. In the clustering stage, we aim to investigate how different model classes can be differentiated using ordinal pattern–based features. Visual inspection of the HC plane will also be used to support the interpretation of the clustering results. In addition, silhouette coefficients will be calculated to assess the quality of the clustering obtained using the K-means algorithm.

2 Methodology

In this experiment, we consider stochastic, deterministic, and chaotic functions. We then couple these models to examine how the points are displayed in the entropy–complexity plane. Below, we describe the simulation setup for our experiment.

Simulation and Feature Construction:

- **Embedding:** A fixed embedding dimension ($D = 4$) and time delay ($\tau = 1$) are used.
- **Model structure:** AR(2), MA(2), ARMA(2,2), a sine function, and the logistic map are considered.
- **Simulation:** For each model and parameter set, $R = 50$ time series of length $n = 1000$ are simulated.
- **Features:** For each realization, the following quantities are computed:
  - *Entropy:* Shannon entropy.
  - *Complexity:* Statistical complexity measures corresponding to Shannon entropy.

In addition to this setup, we will compute Rényi and Tsallis entropies and the Fisher information measure, as well as the associated statistical complexity measures for each entropy type. Moreover, variances will be computed for each model to assess the stability of the features. Confidence intervals for entropy and complexity will also be computed for each model to further assess stability. Once we have examined the spread of the points in the entropy–complexity plane, we will proceed with model validation and time series clustering.

The parameters for these models are summarized in Table 1. The experimental setup includes the following components:

1. Three ARMA-type models:
   - $X_{\text{ARMA}} = \text{ARMA}(2, 2)$
   - $X_{\text{AR}} = \text{AR}(2)$
   - $X_{\text{MA}} = \text{MA}(2)$

Table 1: Parameter settings for the ARMA models used in the simulation study.

Model	**AR coefficients**	**MA coefficients**
ARMA(2, 2)	$(-0.8,\ 0.1)$	$(-0.8,\ 0.1)$
AR(2)	$(0.1,\ 0.8)$	–
MA(2)	–	$(0.1,\ -0.8)$

2. A chaotic logistic map model defined as

$$L_t = rL_{t-1}(1 - L_{t-1}),$$

where $r = 3.8$ is used in the experimental setup, with initial value $L_0 = 0.5$.

3. A sine function defined as

$$f(t) = \sin(2\pi f t),$$

where f and t denote the frequency and time, respectively. For the experimental setup, the frequency is set to $f = 0.05$.

We then construct convex combinations of these components, mirroring the simulation design:

4. Convex combinations of the ARMA$(2, 2)$ process with the logistic map and sine function:

$$X_t^{(\text{ARMA+Logistic})}(w) = w\, X_{\text{ARMA},t} + (1 - w)\, L_t,$$
$$X_t^{(\text{ARMA+Sine})}(w) = w\, X_{\text{ARMA},t} + (1 - w)\, f(t),$$

where $w \in \{0.1, 0.2, \ldots, 0.9\}$.

5. Convex combinations of the AR(2) process with the logistic map and sine function:

$$X_t^{(\text{AR2+Logistic})}(w) = w\, X_{\text{AR},t} + (1 - w)\, L_t,$$

$$X_t^{(\text{AR2+Sine})}(w) = w\, X_{\text{AR},t} + (1 - w)\, f(t).$$

6. Convex combinations of the MA(2) process with the logistic map and sine function:

$$X_t^{(\text{MA2+Logistic})}(w) = w\, X_{\text{MA},t} + (1 - w)\, L_t,$$

$$X_t^{(\text{MA2+Sine})}(w) = w\, X_{\text{MA},t} + (1 - w)\, f(t),$$

again with $w \in \{0.1, 0.2, \ldots, 0.9\}$.

For each model and each weight value of w, we simulate time series of length $n = 1000$ and compute the corresponding entropy and complexity measures.

The experimental setup uses an embedding dimension of $D = 4$. All simulations are generated using the `Mersenne_Twister` pseudorandom number generator.

3 Results and Discussion

The results were obtained based on our experimental setup as follows.

1. Convex combinations of the ARMA$(2, 2)$ process with the logistic map and sine function.

2. Convex combinations of the AR(2) process with the logistic map and sine function.

3. Convex combinations of the MA(2) process with the logistic map and sine function.

4. All models combined together

The Figure 2 shows the results obtained using ARMA$(2, 2)$ with Logistic function. Scatter plot for convex combination with 50 replication is shown in Figure 3

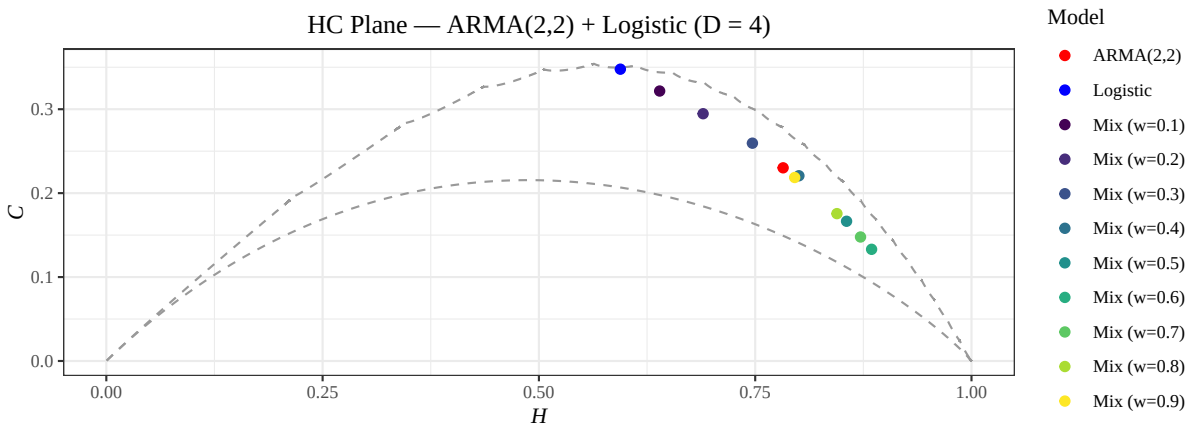

Figure 2: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

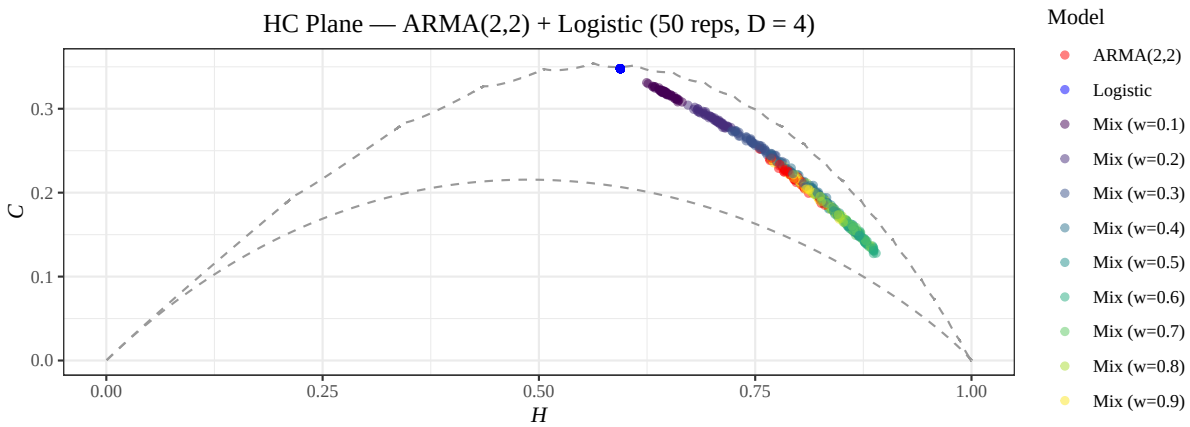

Figure 3: Scatter plot for convex combination model with 50 replications, embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

Table 2: Representative entropy–complexity values for ARMA$(2, 2)$–Logistic mixtures

Model	H **(Entropy)**	C **(Complexity)**
ARMA(2,2)	0.782	0.230
Logistic	0.594	0.348
Mix $(w = 0.1)$	0.640	0.322
Mix $(w = 0.2)$	0.690	0.295
Mix $(w = 0.3)$	0.747	0.260
Mix $(w = 0.4)$	0.801	0.221
Mix $(w = 0.5)$	0.856	0.166
Mix $(w = 0.6)$	0.885	0.133
Mix $(w = 0.7)$	0.872	0.148
Mix $(w = 0.8)$	0.844	0.176
Mix $(w = 0.9)$	0.796	0.219

Referring to the Figure 2, in the HC plane, the points corresponding to the ARMA$(2,2)$–logistic mixtures (convex combinations) mainly lie on the right-hand side of the plot, indicating relatively high entropy and moderate to low complexity. The Table 2 shows the mixed model starts at $w = 0$, which corresponds to the pure logistic process with low entropy (0.594) and high complexity(0.348). As w increases from 0 to 0.3, the complexity decreases further, while the entropy remains relatively high. For larger weights, an interesting pattern emerges: the mixtures with $w = 0.4$ and $w = 0.9$ exhibit entropy and complexity values closer to those of the pure ARMA$(2,2)$ model ($w = 1$), and their entropy even exceeds that of the pure ARMA$(2,2)$ model. The points corresponding to intermediate weights $w = 0.5, 0.6, 0.7$, and 0.8 show consecutively higher entropy and lower complexity. This behaviour illustrates how the choice of weight in the convex combination substantially affects the entropy–complexity characteristics of the ARMA–logistic model.

Following this we make another convex combination model ARMA$(2,2)$ with deterministic sine function. The Figure 4 shows the results obtained using ARMA$(2,2)$ with Sine function. Scatter plot for convex combination with 50 replication is shown in Figure 5.

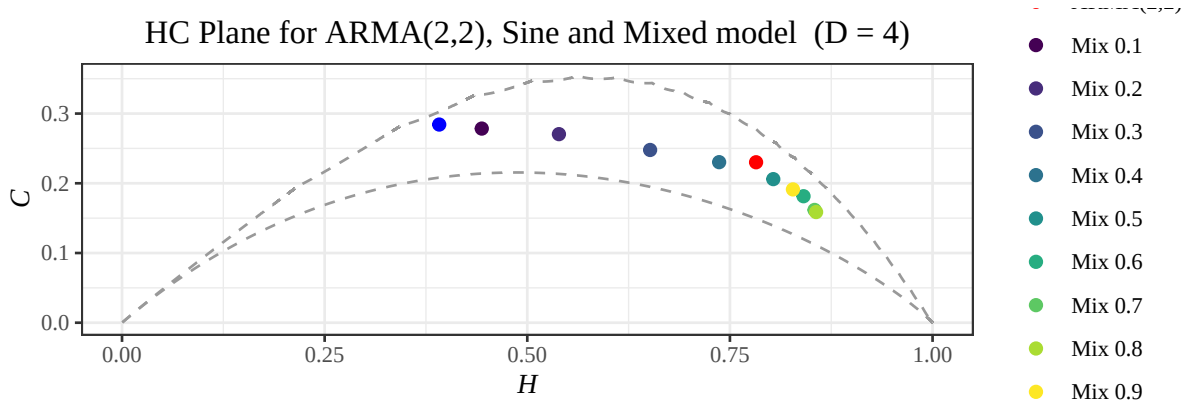

Figure 4: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

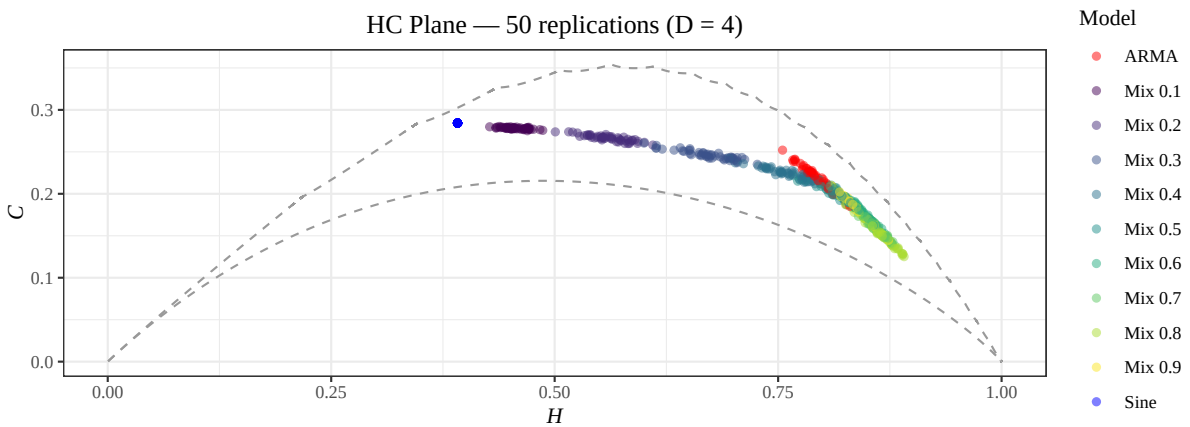

Figure 5: Scatter plot for convex combination model with 50 replications, embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

This convex model gives entropy and complexity measures as shown in the Table 3

Table 3: Representative entropy–complexity values for ARMA$(2,2)$–Sine mixtures

Model	H **(Entropy)**	C **(Complexity)**
ARMA$(2,2)$	0.782	0.230
Sine	0.391	0.284
Mix $w = 0.1$	0.444	0.278
Mix $w = 0.2$	0.539	0.270
Mix $w = 0.3$	0.652	0.248
Mix $w = 0.4$	0.737	0.230
Mix $w = 0.5$	0.803	0.206
Mix $w = 0.6$	0.841	0.181
Mix $w = 0.7$	0.854	0.162
Mix $w = 0.8$	0.856	0.159
Mix $w = 0.9$	0.828	0.191

Table 3 shows that the pure sine model ($w = 0$) has the highest complexity ($C = 0.284$) and the lowest entropy ($H = 0.391$). As the weight w increases, the entropy also increases. Notably, at $w = 0.4$, the complexity ($C = 0.230$) matches that of the pure ARM$A(2,2)$ model ($w = 1$), while the entropy ($H = 0.737$) is substantially higher than that of the pure ARMA model ($H = 0.782$).

A change in pattern begins at $w = 0.5$, where the entropy ($H = 0.803$) is lower than at $w = 0.9$ ($H = 0.828$), but the complexity ($C = 0.206$) is higher than at $w = 0.9$ ($C = 0.191$). From $w = 0.6$ to $w = 0.8$, the entropy continues to increase (from $H = 0.841$ to $H = 0.856$), while the complexity decreases (from $C = 0.181$ to $C = 0.159$). This results also confirm with the Figure 4.

In addition to these convex combination models, from Figure 6 to Figure 13 shows the results obtained using AR(2), MA(2) mixing with logistic map and Sine function.

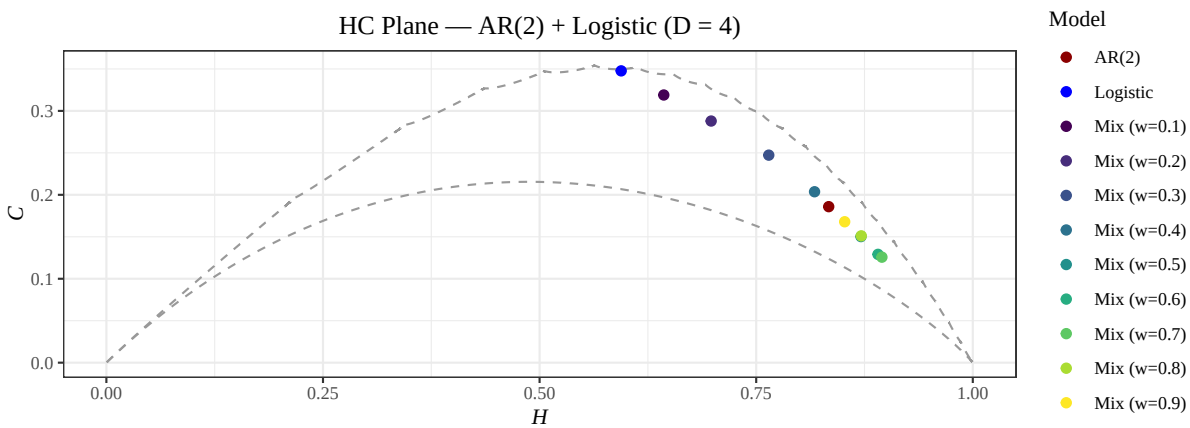

Figure 6: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

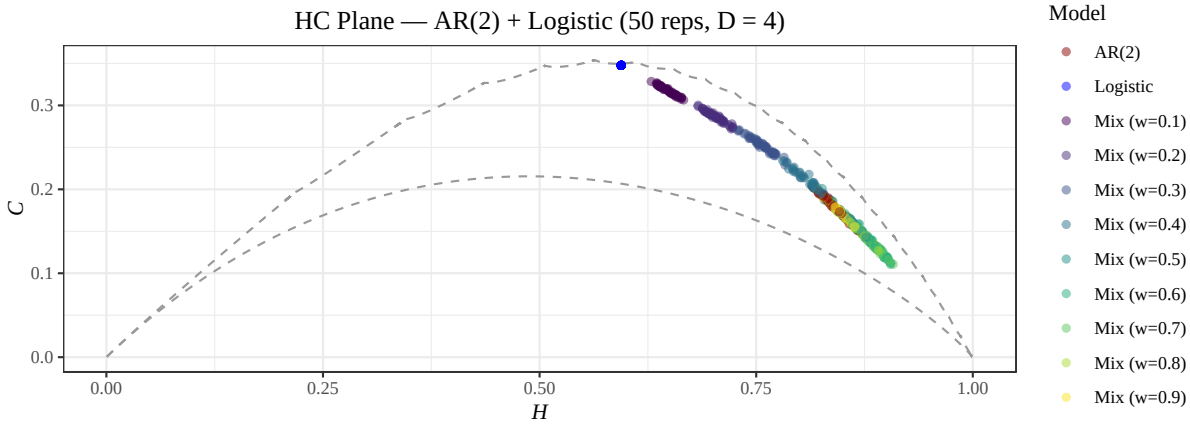

Figure 7: Scatter plot for convex combination model with 50 replications, embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

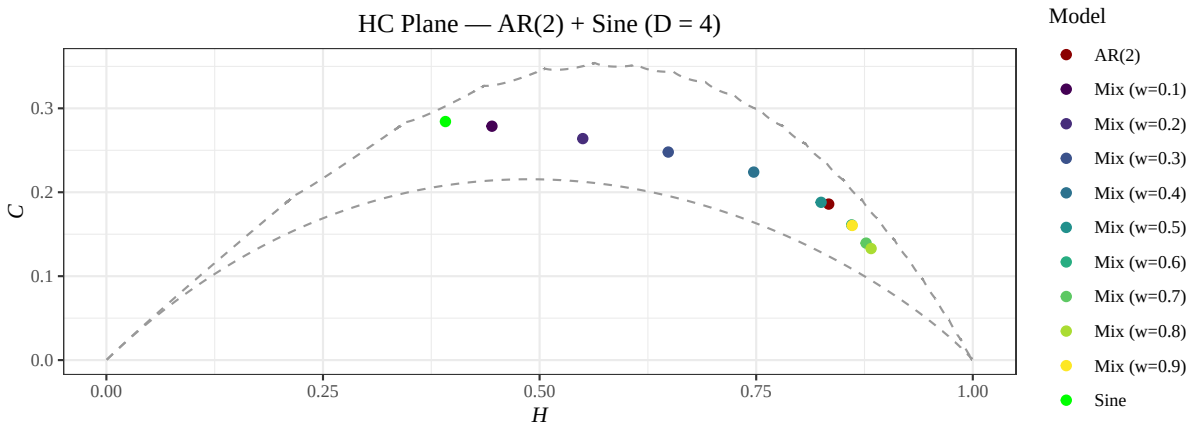

Figure 8: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

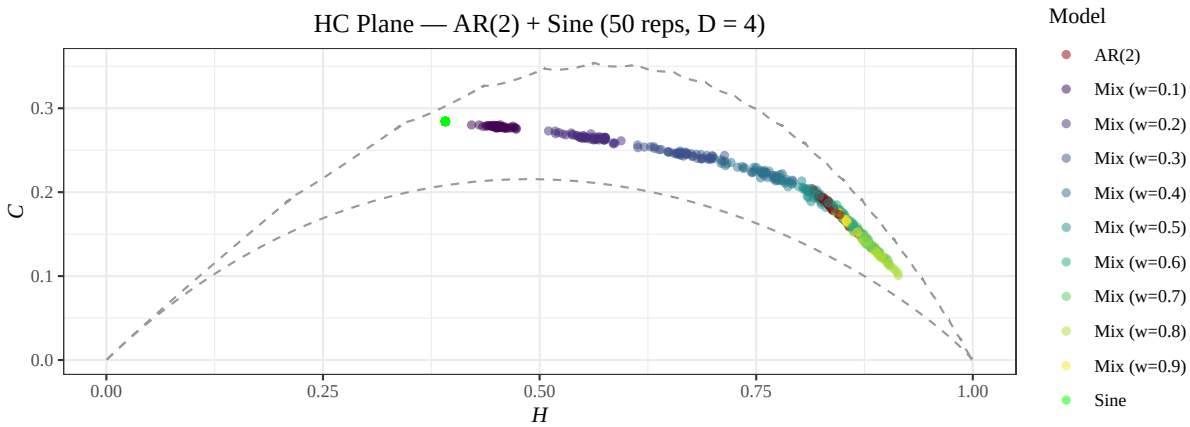

Figure 9: Scatter plot for convex combination model with 50 replications, embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

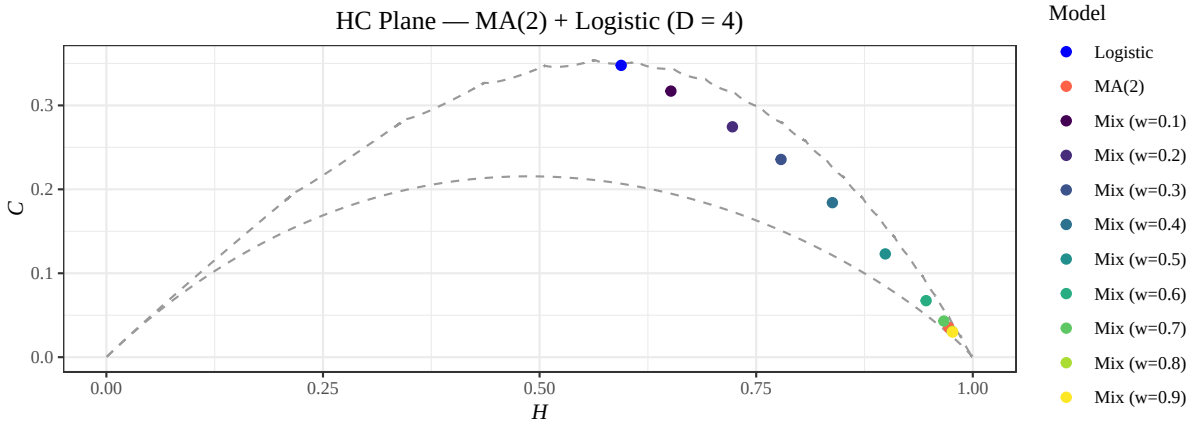

Figure 10: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

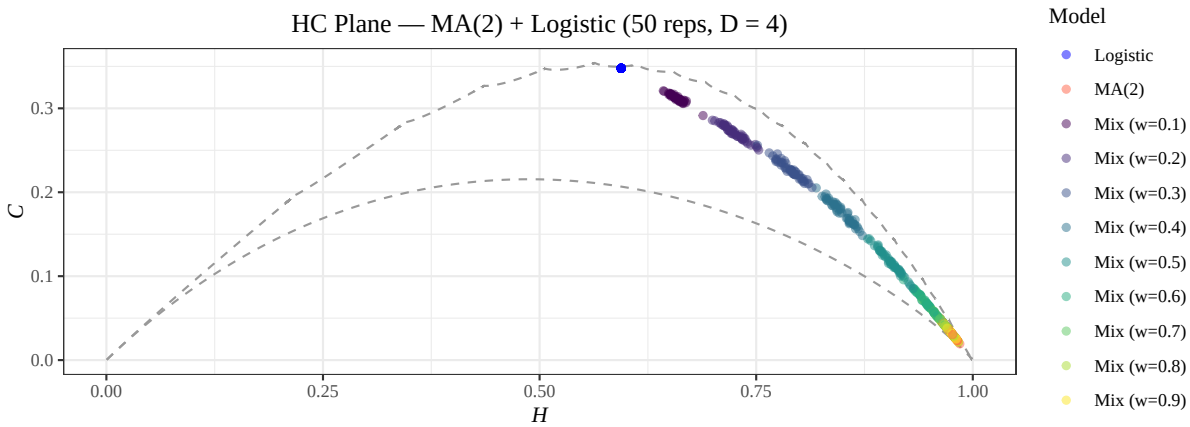

Figure 11: Scatter plot for convex combination model with 50 replications, embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

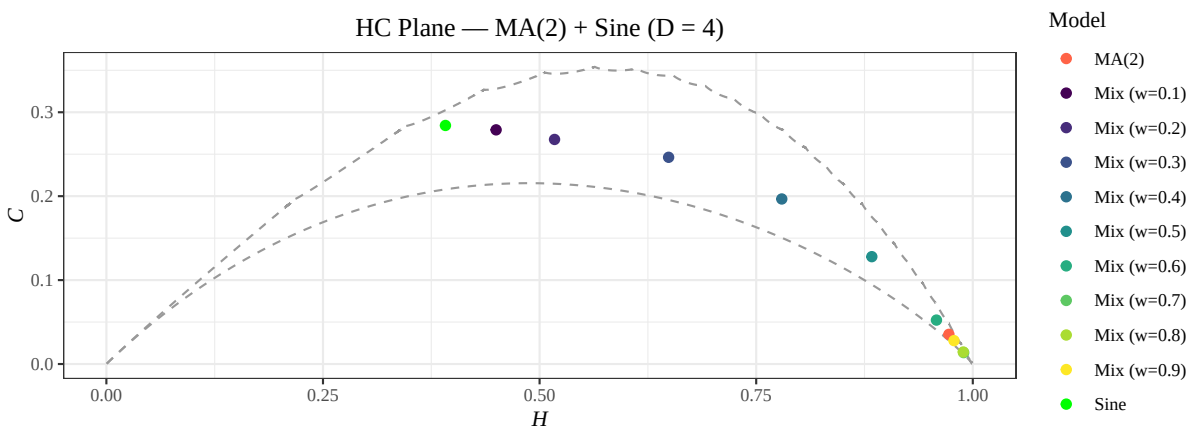

Figure 12: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

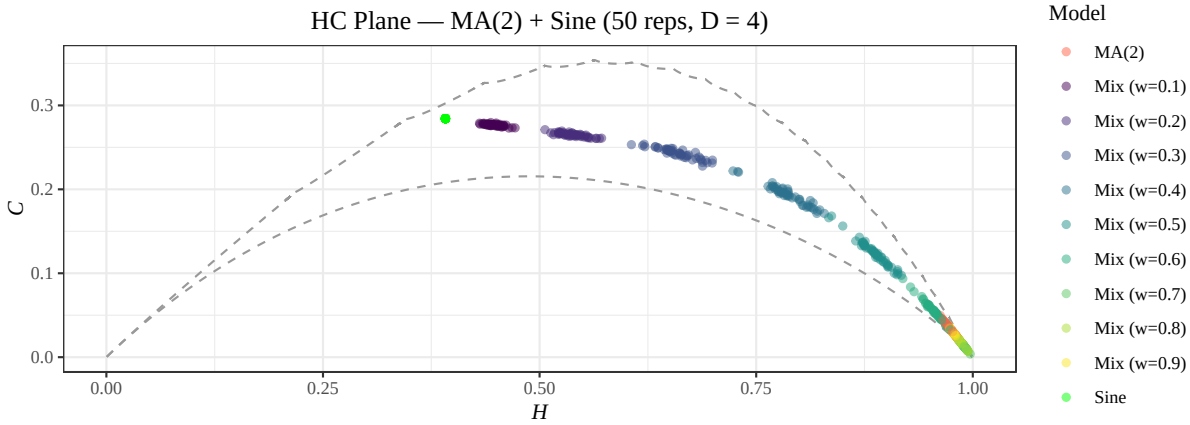

Figure 13: Scatter plot for convex combination model with 50 replications, embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

These figures follow the same pattern described above, with entropy and complexity values exhibiting distinct behaviour at weights $w = 0.5$ and $w = 0.9$. Therefore, I selected the HC plane, which effectively captures all the points across the entropy–complexity space when combining all models together (Figure 14). However, it is difficult to distinguish individual convex combinations in this crowded view (as it shows 9 weights points from each model, for example, AR(2)+Sine show 9 HC points in the plane), so I decided not to include all models in my experimental design. Instead, I chose to focus on combinations involving the MA(2), Sine, and Logistic convex combination models for further investigation.

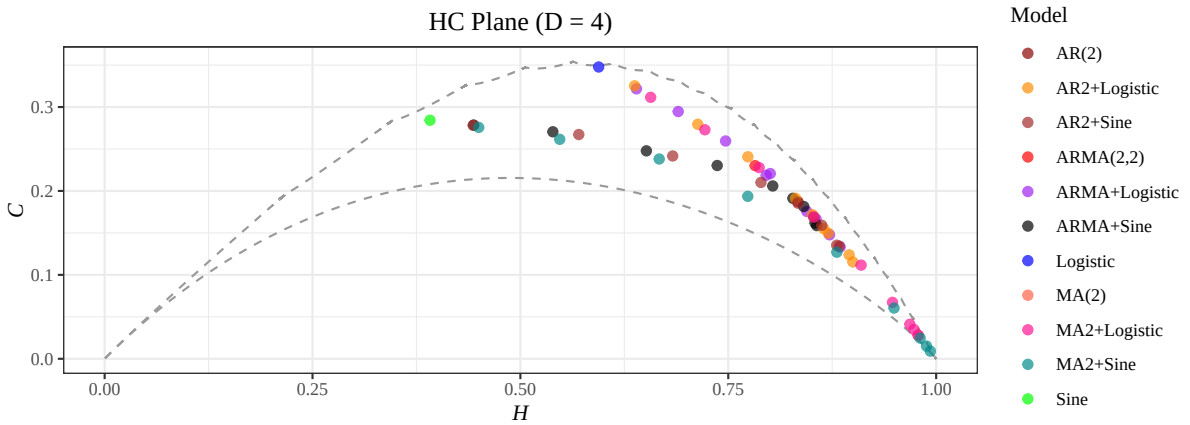

Figure 14: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

The results in Table 4 demonstrate a clear structure in the entropy–complexity plane for the new experimental setup. The points spread nicely across the HC plane and are readily distinguishable. The dataset includes points with both high entropy and low complexity (e.g., MA(2) at $H = 0.972$, $C = 0.035$) as well as low entropy and high complexity (e.g., logistic map at $H = 0.594$, $C = 0.348$).

Notably, the entropy ranges from a maximum of $H = 0.991$ for the MA2+Sine convex combination at $w = 0.6$ down to $H = 0.391$ for the pure sine function ($w = 0$). Similarly, the complexity ranges from a minimum of $C = 0.011$ for MA2+Sine ($w = 0.6$) to a maximum of $C = 0.348$ for the logistic map.

Table 4: Representative entropy–complexity values for MA(2), sine, logistic, and their convex combinations

Model	**Type**	**H (Entropy)**	**C (Complexity)**
MA(2)	Pure	0.972	0.035
Sine	Pure	0.391	0.284
Logistic	Pure	0.594	0.348
MA2+Sine ($w = 0.1$)	MA2+Sine	0.505	0.270
MA2+Sine ($w = 0.2$)	MA2+Sine	0.697	0.234
MA2+Sine ($w = 0.3$)	MA2+Sine	0.847	0.154
MA2+Sine ($w = 0.4$)	MA2+Sine	0.938	0.075
MA2+Sine ($w = 0.5$)	MA2+Sine	0.983	0.021
MA2+Sine ($w = 0.6$)	MA2+Sine	0.991	0.011
MA2+Sine ($w = 0.7$)	MA2+Sine	0.986	0.019
MA2+Sine ($w = 0.8$)	MA2+Sine	0.979	0.027
MA2+Sine ($w = 0.9$)	MA2+Sine	0.973	0.035
MA2+Log ($w = 0.1$)	MA2+Log	0.652	0.317
MA2+Log ($w = 0.2$)	MA2+Log	0.723	0.274
MA2+Log ($w = 0.3$)	MA2+Log	0.779	0.236
MA2+Log ($w = 0.4$)	MA2+Log	0.838	0.184
MA2+Log ($w = 0.5$)	MA2+Log	0.899	0.123
MA2+Log ($w = 0.6$)	MA2+Log	0.946	0.067
MA2+Log ($w = 0.7$)	MA2+Log	0.967	0.043
MA2+Log ($w = 0.8$)	MA2+Log	0.976	0.031
MA2+Log ($w = 0.9$)	MA2+Log	0.977	0.030

It is clear with the Figure 15 and data points are spread around the HC plane as described in the Table 4.

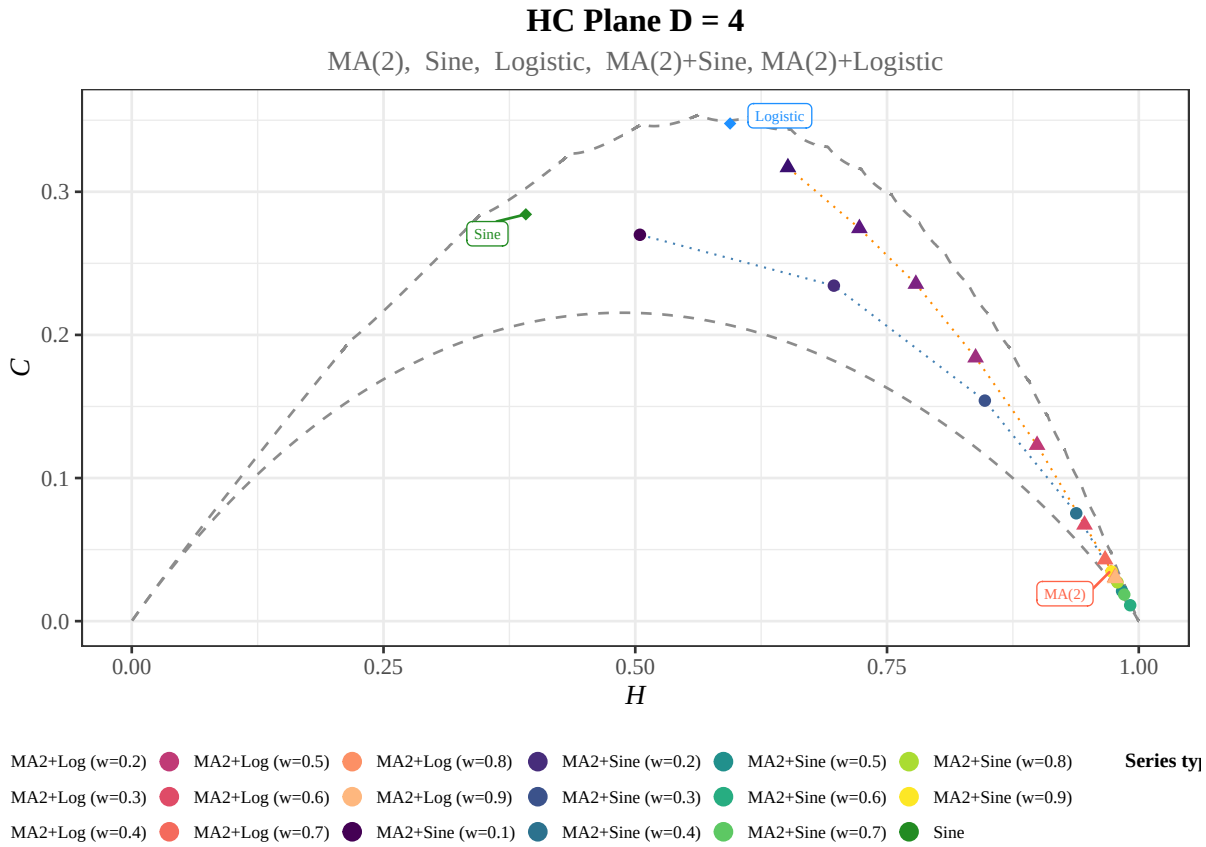

Figure 15: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

Now I will design my future experimental setup based on these simulation results. I will adopt the new simulation configuration shown in the table, which explores a comprehensive range of convex combinations across different time series models and weight values, enabling clear separation of points in the HC plane.

Table 5: Entropy–Complexity values for simulated time series models.

Model	**Type**	**H (Entropy)**	**C (Complexity)**
ARMA(2,2)	Pure	0.8993783	0.12490600
AR(2)	Pure	0.9616821	0.04878726
MA(2)	Pure	0.9760429	0.03092143
Logistic	Pure	0.7422328	0.29230169
Sine	Pure	0.3913538	0.28425599
ARMA+Logistic(w=0.1)	ARMA+Logistic	0.7674692	0.26767468
ARMA+Logistic(w=0.2)	ARMA+Logistic	0.8095423	0.22313552
ARMA+Logistic(w=0.4)	ARMA+Logistic	0.8851289	0.14663422
ARMA+Logistic(w=0.7)	ARMA+Logistic	0.9862352	0.01797356
ARMA+Logistic(w=0.9)	ARMA+Logistic	0.9236793	0.09509173
ARMA+Sine(w=0.1)	ARMA+Sine	0.4484795	0.27481052
ARMA+Sine(w=0.2)	ARMA+Sine	0.5658237	0.26877572
ARMA+Sine(w=0.3)	ARMA+Sine	0.7250210	0.22208325
ARMA+Sine(w=0.4)	ARMA+Sine	0.8195294	0.17747148
ARMA+Sine(w=0.7)	ARMA+Sine	0.8961243	0.12527093
ARMA+Sine(w=0.9)	ARMA+Sine	0.8956125	0.12827562
AR2+Logistic(w=0.2)	AR2+Logistic	0.8132889	0.21879192
AR2+Logistic(w=0.3)	AR2+Logistic	0.8582879	0.17358643
AR2+Logistic(w=0.4)	AR2+Logistic	0.8910653	0.13631995
AR2+Logistic(w=0.7)	AR2+Logistic	0.9816001	0.02422982
AR2+Logistic(w=0.8)	AR2+Logistic	0.9769657	0.03031588
AR2+Logistic(w=0.9)	AR2+Logistic	0.9623328	0.04858689
AR2+Sine(w=0.2)	AR2+Sine	0.5705273	0.26396296
AR2+Sine(w=0.3)	AR2+Sine	0.7133602	0.22249293
AR2+Sine(w=0.4)	AR2+Sine	0.8445838	0.15108954
AR2+Sine(w=0.7)	AR2+Sine	0.9516522	0.06039158
MA2+Logistic(w=0.1)	MA2+Logistic	0.7759295	0.25462025
MA2+Logistic(w=0.2)	MA2+Logistic	0.8309979	0.20119958
MA2+Logistic(w=0.3)	MA2+Logistic	0.8713929	0.15833114
MA2+Logistic(w=0.4)	MA2+Logistic	0.9064925	0.11892669
MA2+Logistic(w=0.5)	MA2+Logistic	0.9498924	0.06464961
MA2+Logistic(w=0.7)	MA2+Logistic	0.9903786	0.01274909
MA2+Sine(w=0.1)	MA2+Sine	0.4965979	0.26996937
MA2+Sine(w=0.2)	MA2+Sine	0.6860130	0.23842479
MA2+Sine(w=0.3)	MA2+Sine	0.8520771	0.15411065
MA2+Sine(w=0.4)	MA2+Sine	0.9360690	0.07548932
MA2+Sine(w=0.5)	MA2+Sine	0.9804243	0.02500202

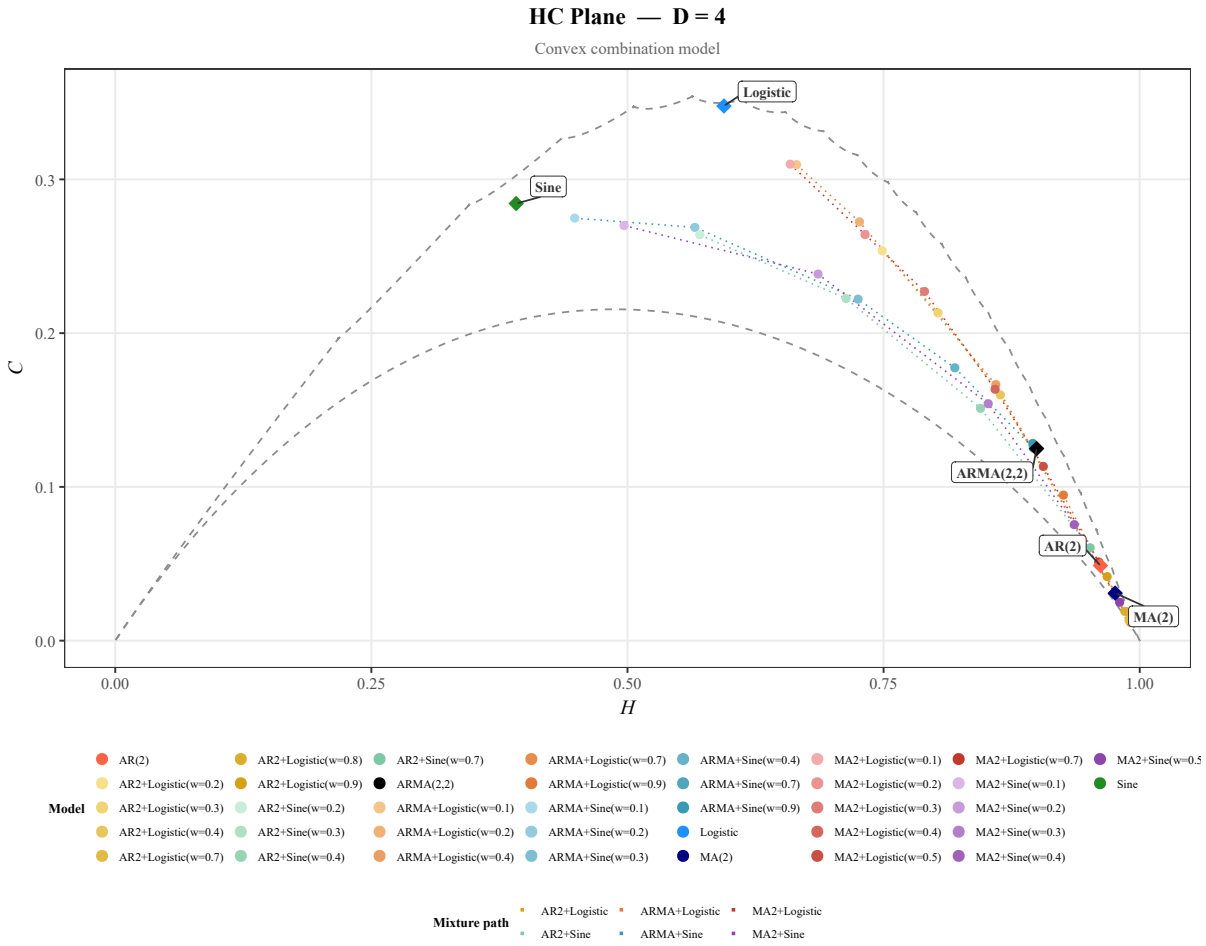

Figure 16: Scatter plot for convex combination model with embedding dimension $D = 4$, time delay $\tau = 1$, and sequence length 1000.

As shown in Figure 16, some entropy–complexity points overlap, making them difficult to distinguish. Therefore, I have excluded these overlapping combinations from my future experimental design. The final experimental setup is based on this refined selection.

References